



# Engineering Exploration

About Computer Science  
and Engineering



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# Intro

- *What engineering is*
- *How it differs from science*
- *Needs and wants of computer science engineer*
- *A case study of how it is implemented in real world projects by combining with other disciplines of engineering*

# What Is Engineering?

Engineering is the application of scientific and mathematical knowledge to solve real, practical problems. It is not theory alone — it is using that knowledge to design, build, test, and improve something that works, under real constraints of cost, time, and safety.



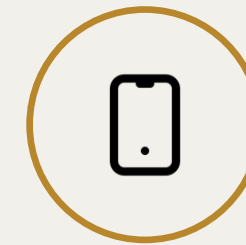
## *The Golden Gate Bridge*

Civil & mechanical engineering solved 4,200 ft of wind-load and suspension-cable math.



## *mRNA Vaccines*

Bioengineering turned Katalin Karikó's lab discovery into a manufacturable process at Pfizer & Moderna.



## *Mobile Phone*

Electrical & computer engineering compressed a 1980s room-sized computer into your pocket.

# The Engineering Method

*Five recurring steps, from question to working solution.*



## OBSERVE

Notice a real problem worth solving



## HYPOTHESIZE

Propose a possible fix



## DESIGN

Work out how to build it



## BUILD

Construct a working version



## TEST & ITERATE

Measure against reality, refine

EX

### REAL-WORLD EXAMPLE – *Wireless Earbuds*

Observe: Sweat causes dropped connections during workouts. Hypothesize: A nano-coating could seal the circuitry. Design & build: Engineers add IPX4 waterproofing. Test & iterate: Thousands of gym users stress-test it before the next model ships.

# Science vs. Engineering

*A single spectrum, not two separate worlds.*



"Why does this happen?" It studies nature's laws to generate knowledge — often without an immediate practical goal.



"How can I build something useful?" It applies known laws to design and ship a working solution.

EX

## REAL-WORLD EXAMPLE — GPS & Google Maps

Physicists proved that time runs faster for satellites in orbit. Engineers used that exact correction to keep GPS clocks accurate — without it, Google Maps would drift by about 10 km per day.

# The Scientist vs. The Engineer

*Two very different working days.*



## A SCIENTIST'S DAY

- Observe — notices an anomaly
- Hypothesize — proposes an explanation
- Experiment — runs controlled tests
- Publish — shares findings to verify



## AN ENGINEER'S DAY

- Define — scopes a real problem to solve
- Design — plans a working solution
- Build — constructs and assembles it
- Ship & Test — deploys, measures results

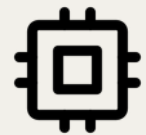
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### REAL-WORLD EXAMPLE — *Katalin Karikó*

As a scientist, she discovered mRNA could evade immune detection. Engineers at Moderna and Pfizer–BioNTech turned that discovery into a manufacturable vaccine platform used by billions.

# Why CS Engineering Is Rising

*Five forces reshaping the discipline.*



## Everything Is Becoming Software

Cars, appliances, and factories increasingly run on code.



## The AI & Data Explosion

ML needs engineers who can build & deploy models at scale.



## Cloud & Cybersecurity Demand

Every organization now depends on secure, always-on infrastructure.



## Portable, Global Work

Software roles are among the most flexible careers today.



## Constant Reinvention

New frameworks & chips create fresh specializations yearly.

EX

### REAL-WORLD EXAMPLE — *100 Million Lines of Code*

A single modern Tesla runs more lines of code than an F-35 fighter jet — most of it written and maintained by CS engineers.

# Needs & Wants of a CS Engineer

*A foundation of needs, topped by what makes the work fulfilling.*

## FOUNDATION — NEEDS

- ✓ Programming, data structures & algorithms
- ✓ Understanding hardware, networks & data
- ✓ Adaptability, since technology shifts fast

## ASPIRATION — WANTS

- ✓ Meaningful problems, not just code
- ✓ Continuous learning & collaboration
- ✓ Cutting-edge work: AI, security, cloud

EX

### REAL-WORLD EXAMPLE — *The Friday-Night Order Spike*

An engineer at a food-delivery company needs AWS cloud fluency to keep the app running when orders triple overnight — and wants the chance to design the routing algorithm that decides which driver gets which order.

# Various Disciplines of Engineering

- Computer Science Engineering (CSE)
- Electronics & Communication (ECE)
- Electrical Engineering (EEE)
- Mechanical Engineering
- Civil Engineering
- Agricultural Engineering
- Instrumentation Engineering

# Case Study — The Tesla Model 3

*One real, mass-produced electric vehicle, engineered by six disciplines at once. The next four plates take it apart, piece by piece.*



TESLA MODEL 3 • FULL SELF-DRIVING HARDWARE



SENSORS



POWER



SOFTWARE



SAFETY



MFG.



INFRA.

# Branches Involved

## Mechanical & Electrical Systems



### MECHANICAL ENGINEERING

- Chassis & crash structure absorb impact energy to protect occupants.
- Suspension & steering tuned for ride comfort and precise autonomous control.
- Brake-by-wire replaces mechanical linkages with redundant, electronic actuation.



### ELECTRICAL ENGINEERING

- 8 exterior cameras & 12 ultrasonic sensors feed the perception system.
- Forward radar adds all-weather object detection cameras alone can miss.
- A 60–100 kWh battery pack powers both propulsion and onboard compute.

# Civil & Industrial Systems

## CIVIL ENGINEERING

- HD maps encode lane geometry & signal positions to centimeter accuracy.
- GPS & inertial sensors cross-reference position against those maps.
- Road design standards directly shape what the software must handle.

## INDUSTRIAL ENGINEERING

- Robotic arms handle welding, painting & precision assembly on the line.
- Statistical quality control catches defects before a car leaves the plant.
- Just-in-time supply chains coordinate thousands of parts arriving on cue.

# Computer Science & Safety Systems

## COMPUTER SCIENCE ENGINEERING

- Neural nets trained on millions of miles learn to recognize pedestrians & signs.
- Sensor-fusion combines camera, radar & ultrasonic data into one live world-model.
- Path-planning decides how to steer, brake & accelerate, many times per second.
- Over-the-air updates improve the whole fleet after purchase — the connective layer.

## SAFETY & SYSTEMS ENGINEERING

- Redundant braking & steering paths mean no single failure disables the car.
- The system fails safely, degrading gracefully rather than shutting off.
- Simulation runs millions of virtual test miles nightly before code reaches a car.

# The Tesla Model 3, By the Numbers

*Representative specifications for camera-and-radar autonomous driving hardware.*

8+

EXTERIOR CAMERAS

12

ULTRASONIC SENSORS

1

FORWARD RADAR UNIT

60–100 kWh

BATTERY CAPACITY

Millions

TRAINING MILES (REAL + SIM)

ISO 26262

FUNCTIONAL-SAFETY STANDARD

# Misconceptions About CS Engineers

- |                                              |                                                                                   |
|----------------------------------------------|-----------------------------------------------------------------------------------|
| ⊗ <i>"It's only about coding"</i>            | ✓ Coding is one tool — the real work is problem-solving & system design.          |
| ⊗ <i>"They work alone"</i>                   | ✓ Most software is built by large, collaborative teams.                           |
| ⊗ <i>"Learn one language, done for life"</i> | ✓ The field demands constant upskilling as tools evolve.                          |
| ⊗ <i>"Only for math geniuses"</i>            | ✓ Logical thinking & consistent practice matter more than raw talent.             |
| ⊗ <i>"AI will replace programmers"</i>       | ✓ AI is a tool engineers use — someone still designs, verifies & owns the system. |

# Skills for the 21st Century CS Engineer



## Technical Core

Programming, algorithms, databases (e.g., building a REST API)



## Collaboration

Teamwork, explaining ideas simply (e.g., a clear pull-request review)



## Ethical Mindset

Responsible use of data & systems (e.g., auditing a hiring algorithm for bias)



## AI & Cloud

ML basics, AWS/Azure, cybersecurity (e.g., deploying a model to production)



## Lifelong Learning

Constant upskilling as tools evolve (e.g., learning Rust after C++)



## Problem-Solving

Adapting to unfamiliar challenges (e.g., debugging a live production outage)

*Together, these six capabilities form the working core of the modern CS engineer.*

# THANK YOU

